Week 7: SQL Query Implementation for E-Commerce Database

The e-commerce company has collected customer, product, order, and inventory data. The management team wants to retrieve useful information from the database for daily operations and basic business analysis.

Perform Basic SQL Queries

1. Display all customer details.

```
SQL> SELECT * FROM CUSTOMER;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                
```

```
        13 Akash Kumar
akash@gmail.com                                                    9078237684
Chennai
16-JUL-25

        14 Gayathri R
gayu@gmail.com                                                     8964785690
Salem
17-AUG-25

        15 Ishitha R
ishitha@gmail.com                                                  9085682356
Vellore
10-SEP-25


15 rows selected.
```

2. Display all available products.

```
SQL> SELECT * FROM PRODUCT;

PRODUCT PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
P001    Smart phone                    1      24999             50 S001
P002    Wireless Mouse                 1        799            100 S002
P003    Bluetooth Speaker              1       1499             75 S003
P004    Cotton Shirt                   2        899             60 S004
P005    Denim Jeans                    2       1599             45 S005
P006    Data Structures                3        650             30 S006
P007    Python programming             3        850             25 S007
P008    Electric Kettle                4       1299             35 S008
P009    Running Shoes                  5       2199             40 S009
P010    Office Chair                   9       5499             15 S010
P011    Building Blocks               10        999             35
P012    Cooking oil                    7        699             70
P013    Face wash                      6        349             90
P014    Study Table                    9       6999             10
P015    Yoga Mat                       5        599             80

15 rows selected.
```

3. Retrieve product names and prices only.

```
SQL> SELECT PRODUCT_NAME, PRICE FROM PRODUCT;

PRODUCT_NAME              PRICE
-------------------- ----------
Smart phone                24999
Wireless Mouse               799
Bluetooth Speaker           1499
Cotton Shirt                 899
Denim Jeans                 1599
Data Structures              650
Python programming           850
Electric Kettle             1299
Running Shoes               2199
Office Chair                5499
Building Blocks              999
Cooking oil                  699
Face wash                    349
Study Table                 6999
Yoga Mat                     599

15 rows selected.
```

4. **Display all orders placed by customers.**

```
SQL> SELECT C.CUSTOMER_NAME,O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID;

CUSTOMER_NAME                                      ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------------------------------------- ------- ----------- --------- ------------ ------------
Arun Kumar                                         O001              1 01-SEP-26         2500 DELIVERED
Priya Sharma                                       O002              2 02-SEP-26         1800 SHIPPED
Arun Kumar                                         O003              1 03-SEP-26         3200 DELIVERED
Rahul Raj                                          O004              3 04-SEP-26         1500 PENDING
Karthik M                                          O005              4 05-SEP-26         4200 DELIVERED
Priya Sharma                                       O006              2 06-SEP-26         2750 SHIPPED
Anjali Devi                                        O007              5 07-SEP-26         3600 DELIVERED
Vijay Kumar                                        O008              6 08-SEP-26         1200 PENDING
Arun Kumar                                         O009              1 09-SEP-26         5100 DELIVERED
Meeena K                                           O010              7 10-SEP-26         2300 SHIPPED
Ajay Kumar                                         O011              8 11-SEP-26         1900 DELIVERED
Rahul Raj                                          O012              3 12-SEP-26         3400 PENDING
Manoj R                                            O013              9 13-SEP-26         2800 SHIPPED
Nithya S                                           O014             10 14-SEP-26         4500 DELIVERED
Karthik M                                          O015              4 15-SEP-26         1600 PENDING

15 rows selected.
```

5. **Retrieve payment details.**

```
SQL> select * from payment;

PAYMENT ORDER_I PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT
------- ------- --------- --------------- ---------- ------------------
PM001   O001    01-SEP-26 Card            Success                2500
PM002   O002    02-SEP-26 UPI             Success                1800
PM003   O003    03-SEP-26 Net Banking     Success                3200
PM004   O004    04-SEP-26 Card            Failed                 1500
PM005   O005    05-SEP-26 UPI             Success                4200
PM006   O006    06-SEP-26 Card            Success                2750
PM007   O007    07-SEP-26 Net Banking     Success                3600
PM008   O008    08-SEP-26 Cash            Pending                1200
PM009   O009    09-SEP-26 UPI             Success                5100
PM010   O010    10-SEP-26 Card            Success                2300
PM011   O011    11-SEP-26 UPI             Success                1900
PM012   O012    12-SEP-26 Net Banking     Pending                3400
PM013   O013    13-SEP-26 Card            Success                2800
PM014   O014    14-SEP-26 UPI             Success                4500
PM015   O015    15-SEP-26 Cash            Pending                1600

15 rows selected.
```

Apply Filtering Conditions Using WHERE

Find products with price greater than ₹5000.

```
SQL> SELECT * FROM PRODUCT WHERE PRICE > 5000;

PRODUCT PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
P001    Smart phone                    1      24999             50 S001
P010    Office Chair                   9       5499             15 S010
P014    Study Table                    9       6999             10
```

Display completed orders.

```
SQL> SELECT * FROM ORDERS WHERE ORDER_STATUS='DELIVERED';

ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------- ----------- --------- ------------ ---------------
O001              1 01-SEP-26         2500 DELIVERED
O003              1 03-SEP-26         3200 DELIVERED
O005              4 05-SEP-26         4200 DELIVERED
O007              5 07-SEP-26         3600 DELIVERED
O009              1 09-SEP-26         5100 DELIVERED
O011              8 11-SEP-26         1900 DELIVERED
O014             10 14-SEP-26         4500 DELIVERED

7 rows selected.
```

Find products with rating above 3.

```
SQL> SELECT P.PRODUCT_NAME,ROUND(AVG(R.RATING),2) AS AVERAGE_RATING FROM PRODUCT P JOIN REVIEW R ON P.PRODUCT_ID=R.P
RODUCT_ID GROUP BY P.PRODUCT_ID,P.PRODUCT_NAME HAVING AVG(R.RATING)>3 ORDER BY AVERAGE_RATING DESC;

PRODUCT_NAME         AVERAGE_RATING
-------------------- --------------
Python programming                5
Smart phone                       5
Face wash                         5
Office Chair                      5
Bluetooth Speaker                 5
Denim Jeans                       4
Wireless Mouse                    4
Yoga Mat                          4
Cooking oil                       4
Running Shoes                     4

10 rows selected.
```

Sort Data Using ORDER BY

Display products from lowest to highest price

```
SQL> SELECT * FROM PRODUCT ORDER BY PRICE;

PRODUCT PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
P013    Face wash                      6        349             90
P015    Yoga Mat                       5        599             80
P006    Data Structures                3        650             30 S006
P012    Cooking oil                    7        699             70
P002    Wireless Mouse                 1        799            100 S002
P007    Python programming             3        850             25 S007
P004    Cotton Shirt                   2        899             60 S004
P011    Building Blocks               10        999             35
P008    Electric Kettle                4       1299             35 S008
P003    Bluetooth Speaker              1       1499             75 S003
P005    Denim Jeans                    2       1599             45 S005

PRODUCT PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
P009    Running Shoes                  5       2199             40 S009
P010    Office Chair                   9       5499             15 S010
P014    Study Table                    9       6999             10
P001    Smart phone                    1      24999             50 S001

15 rows selected.
```

Display customers alphabetically.

```
SQL> SELECT * FROM CUSTOMER ORDER BY CUSTOMER_NAME;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                                PHONE
---------------------------------------------------------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------------------------------------------
REGISTRAT
---------
          8 Ajay Kumar
ajay@gmail.com                                                                                       9876543220
Chennai
10-APR-25

         13 Akash Kumar
akash@gmail.com                                                                                      9078237684
Chennai
16-JUL-25

          5 Anjali Devi
anjali@gmail.com                                                                                     9876543215
Salem
01-MAR-25

          1 Arun Kumar
arun@gmail.com                                                                                       9876543210
Chennai
10-JAN-25

         14 Gayathri R
gayu@gmail.com                                                                                       8964785690
Salem
17-AUG-25

         15 Ishitha R
ishitha@gmail.com                                                                                    9085682356
Vellore
10-SEP-25

          4 Karthik M
karthik@gmail.com                                                                                    9876543214
Chennai
20-FEB-25
```

```
         9 Manoj R
manoj@gmail.com                                                                 9876543222
Tirunelveli
01-MAY-25

         7 Meeena K

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                           PHONE
------------------------------------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------------------------------------------------------
REGISTRAT
---------
meena@gmail.com                                                                 9876543219
Vellore
02-APR-25

        10 Nithya S
nithya@gmail.com                                                                9756864237
Coimbatore
12-MAY-25

         2 Priya Sharma
priya@gmail.com                                                                 9876543211
Bangalore
15-JAN-25

         3 Rahul Raj
rahul@gmail.com                                                                 9876543212
Coimbatore
05-FEB-25

        11 Ravi Chandran
ravi@gmail.com                                                                  47589364470
Pondicherry
20-MAY-25

        12 Suresh Babu
suresh@gmail.com                                                                7895634786
Erode
25-JUN-25

         6 Vijay Kumar
vijay@gmail.com                                                                 9876543216
```

Find top expensive products.

```
SQL> SELECT * FROM PRODUCT WHERE ROWNUM <=5 ORDER BY PRICE DESC;

PRODUCT PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
P001    Smart phone                    1      24999             50 S001
P005    Denim Jeans                    2       1599             45 S005
P003    Bluetooth Speaker              1       1499             75 S003
P004    Cotton Shirt                   2        899             60 S004
P002    Wireless Mouse                 1        799            100 S002
```

Display latest orders first.

```
SQL> SELECT * FROM ORDERS ORDER BY ORDER_DATE DESC;

ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------- ----------- --------- ------------ -------------
O015              4 15-SEP-26         1600 PENDING
O014             10 14-SEP-26         4500 DELIVERED
O013              9 13-SEP-26         2800 SHIPPED
O012              3 12-SEP-26         3400 PENDING
O011              8 11-SEP-26         1900 DELIVERED
O010              7 10-SEP-26         2300 SHIPPED
O009              1 09-SEP-26         5100 DELIVERED
O008              6 08-SEP-26         1200 PENDING
O007              5 07-SEP-26         3600 DELIVERED
O006              2 06-SEP-26         2750 SHIPPED
O005              4 05-SEP-26         4200 DELIVERED
O004              3 04-SEP-26         1500 PENDING
O003              1 03-SEP-26         3200 DELIVERED
O002              2 02-SEP-26         1800 SHIPPED
O001              1 01-SEP-26         2500 DELIVERED

15 rows selected.
```

Retrieve Unique Values Using DISTINCT

Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PAYMENT_MODE FROM PAYMENT;

PAYMENT_MODE
---------------
Net Banking
Card
Cash
UPI
```

Search Products Based on Conditions

Find products within a specific price range. (Product Between 20,000 to 40,000)

```
SQL> SELECT * FROM PRODUCT WHERE PRICE BETWEEN 20000 AND 40000;

PRODUCT PRODUCT_NAME          CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
P001    Smart phone                    1      24999             50 S001
```

Display products belonging to a particular category.

```
SQL> SELECT * FROM PRODUCT WHERE CATEGORY_ID='3';

PRODUCT PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
P006    Data Structures                3        650             30 S006
P007    Python programming             3        850             25 S007
```

Search products using product name.

```
SQL> SELECT * FROM PRODUCT WHERE PRODUCT_NAME='Smart phone';

PRODUCT PRODUCT_NAME          CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
P001    Smart phone                    1      24999             50 S001
```

Find low-stock products.

```
SQL> SELECT * FROM INVENTORY WHERE STOCK_QUANTITY < 6;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATUS         LAST_UPDA
------- ------- ------- -------------- -------------------- ---------
I006    P006    S006                5 Low Stock            17-SEP-26
I008    P008    S008                0 Out of Stock         17-SEP-26
I010    P010    S010                3 Low Stock            17-SEP-26
```

Retrieve Customer and Product Information

Display customer details with their orders.

```
SQL> SELECT C.*,O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID ORDER BY C.CUSTOMER_ID, O.ORDER_DATE;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                                                                PHONE
------------------------------------------------------------------------------------------------------------------------------------ ----------
ADDRESS
--------------------------------------------------------------------------------------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------

          1 Arun Kumar
arun@gmail.com                                                                                                                       9876543210
Chennai
10-JAN-25 O001              1 01-SEP-26         2500 DELIVERED

          1 Arun Kumar
arun@gmail.com                                                                                                                       9876543210
Chennai
10-JAN-25 O003              1 03-SEP-26         3200 DELIVERED

          1 Arun Kumar
arun@gmail.com                                                                                                                       9876543210
Chennai
10-JAN-25 O009              1 09-SEP-26         5100 DELIVERED

          2 Priya Sharma
priya@gmail.com                                                                                                                      9876543211
Bangalore
15-JAN-25 O002              2 02-SEP-26         1800 SHIPPED

          2 Priya Sharma
priya@gmail.com                                                                                                                      9876543211
Bangalore
15-JAN-25 O006              2 06-SEP-26         2750 SHIPPED

          3 Rahul Raj
rahul@gmail.com                                                                                                                      9876543212
Coimbatore
05-FEB-25 O004              3 04-SEP-26         1500 PENDING

          3 Rahul Raj
rahul@gmail.com                                                                                                                      9876543212
Coimbatore
05-FEB-25 O012              3 12-SEP-26         3400 PENDING

          4 Karthik M
karthik@gmail.com                                                                                                                    9876543214
Chennai
20-FEB-25 O005              4 05-SEP-26         4200 DELIVERED

          4 Karthik M
```

```
CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                                                                PHONE
------------------------------------------------------------------------------------------------------------------------------------ ----------
ADDRESS
--------------------------------------------------------------------------------------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------
karthik@gmail.com                                                                                                                    9876543214
Chennai
20-FEB-25 O015              4 15-SEP-26         1600 PENDING

          5 Anjali Devi
anjali@gmail.com                                                                                                                     9876543215
Salem
01-MAR-25 O007              5 07-SEP-26         3600 DELIVERED

          6 Vijay Kumar
vijay@gmail.com                                                                                                                      9876543216
Chennai
08-MAR-25 O008              6 08-SEP-26         1200 PENDING

          7 Meeena K
meena@gmail.com                                                                                                                      9876543219
Vellore
02-APR-25 O010              7 10-SEP-26         2300 SHIPPED

          8 Ajay Kumar
ajay@gmail.com                                                                                                                       9876543220
Chennai
10-APR-25 O011              8 11-SEP-26         1900 DELIVERED

          9 Manoj R
manoj@gmail.com                                                                                                                      9876543222
Tirunelveli
01-MAY-25 O013              9 13-SEP-26         2800 SHIPPED

         10 Nithya S
nithya@gmail.com                                                                                                                     9756864237
Coimbatore
12-MAY-25 O014             10 14-SEP-26         4500 DELIVERED


15 rows selected.
```

Display product details with category information.

```
SQL> SELECT C.*, P.* FROM PRODUCT P JOIN CATEGORY C ON P.CATEGORY_ID=C.CATEGORY_ID;

CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
DESCRIPTION
--------------------------------------------------------------------------------
PRODUCT PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------
          1 Electronics
Electronic devices and accessories
P001    Smart phone                   1      24999             50 S001

          1 Electronics
Electronic devices and accessories
P002    Wireless Mouse                1        799            100 S002

          1 Electronics
Electronic devices and accessories
P003    Bluetooth Speaker             1       1499             75 S003

          2 Clothing
Men and women clothing products
P004    Cotton Shirt                  2        899             60 S004

          2 Clothing
Men and women clothing products
P005    Denim Jeans                   2       1599             45 S005

          3 Books
Educational and general books
P006    Data Structures               3        650             30 S006

          3 Books
Educational and general books
P007    Python programming            3        850             25 S007

          4 Home Appliances
Appliances used at home
P008    Electric Kettle               4       1299             35 S008

          5 Sports
Sports and fitness equipment
P009    Running Shoes                 5       2199             40 S009

          9 Furniture
Home and office furniture
P010    Office Chair                  9       5499             15 S010

         10 Toys
Toys and games for children
P011    Building Blocks              10        999             35

CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
DESCRIPTION
--------------------------------------------------------------------------------
PRODUCT PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY SELLER_
------- -------------------- ----------- ---------- -------------- -------

          7 Grocery
Daily grocery and food products
P012    Cooking oil                   7        699             70

          6 Beauty
Beauty and personal care products
P013    Face wash                     6        349             90

          9 Furniture
Home and office furniture
P014    Study Table                   9       6999             10

          5 Sports
Sports and fitness equipment
P015    Yoga Mat                      5        599             80


15 rows selected.
```

Find customers who purchased a specific product.

```
SQL> SELECT DISTINCT C.CUSTOMER_ID,C.CUSTOMER_NAME,C.EMAIL,C.PHONE FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.
CUSTOMER_ID JOIN ORDER_DETAILS OD ON O.ORDER_ID=OD.ORDER_ID JOIN PRODUCT P ON OD.PRODUCT_ID=P.PRODUCT_ID WHERE P.PRO
DUCT_ID='P011';

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                    PHONE
-------------------------------------------------------------------------------------------------------------- ---------------
          8 Ajay Kumar
ajay@gmail.com                                           9876543220
```

Display products purchased by each customer.

```
SQL> SELECT C.CUSTOMER_ID,C.CUSTOMER_NAME,O.ORDER_ID,P.PRODUCT_ID,OD.QUANTITY,OD.PRICE,OD.QUANTITY * OD.PRICE AS LIN
E_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID JOIN ORDER_DETAILS OD ON O.ORDER_ID=OD.ORDER_I
D JOIN PRODUCT P ON OD.PRODUCT_ID=P.PRODUCT_ID ORDER BY C.CUSTOMER_ID,O.ORDER_DATE,P.PRODUCT_NAME;

CUSTOMER_ID CUSTOMER_NAME                                    ORDER_I PRODUCT    QUANTITY      PRICE LINE_AMOUNT
----------- -------------------------------------------------- ------- ------- ---------- ---------- -----------
          1 Arun Kumar                                         O001    P001             1       2500        2500
          1 Arun Kumar                                         O003    P003             1       3200        3200
          1 Arun Kumar                                         O009    P009             3       1700        5100
          2 Priya Sharma                                       O002    P002             2        900        1800
          2 Priya Sharma                                       O006    P006             1       2750        2750
          3 Rahul Raj                                          O004    P004             3        500        1500
          3 Rahul Raj                                          O012    P012             1       3400        3400
          4 Karthik M                                          O005    P005             2       2100        4200
          4 Karthik M                                          O015    P015             2        800        1600
          5 Anjali Devi                                        O007    P007             2       1800        3600
          6 Vijay Kumar                                        O008    P008             1       1200        1200
          7 Meeena K                                           O010    P010             1       2300        2300
          8 Ajay Kumar                                         O011    P011             2        950        1900
          9 Manoj R                                            O013    P013             1       4500        4500
         10 Nithya S                                           O014    P014             1       4500        4500

15 rows selected.
```

Apply Multiple Filtering Conditions

Find electronics products costing more than ₹10000.

```
SQL> SELECT P.PRODUCT_ID,P.PRODUCT_NAME,C.CATEGORY_NAME, P.PRICE FROM PRODUCT P JOIN CATEGORY C ON P.CATEGORY_ID=C.C
ATEGORY_ID WHERE PRICE> 10000 ORDER BY P.PRICE DESC;

PRODUCT PRODUCT_NAME         CATEGORY_NAME                                      PRICE
------- -------------------- -------------------------------------------------- ----------
P001    Smart phone          Electronics                                        24999
```

Search products containing the word MOBILE".

```
SQL> SELECT PRODUCT_ID,PRODUCT_NAME, PRICE, STOCK_QUANTI
TY FROM PRODUCT WHERE UPPER(PRODUCT_NAME)LIKE '%mobile';


no rows selected
```

Find orders placed within a specific date range.

```
SQL> SELECT * FROM ORDERS WHERE ORDER_DATE BETWEEN DATE
'2025-09-01' AND DATE '2026-09-10' ORDER BY ORDER_DATE;

ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------- ----------- --------- ------------ -------------
-------
0001                1 01-SEP-26         2500 DELIVERED
0002                2 02-SEP-26         1800 SHIPPED
0003                1 03-SEP-26         3200 DELIVERED
0004                3 04-SEP-26         1500 PENDING
0005                4 05-SEP-26         4200 DELIVERED
0006                2 06-SEP-26         2750 SHIPPED
0007                5 07-SEP-26         3600 DELIVERED
0008                6 08-SEP-26         1200 PENDING
0009                1 09-SEP-26         5100 DELIVERED
0010                7 10-SEP-26         2300 SHIPPED

10 rows selected.
```

Generate Basic Business Reports

Report 1: Product Availability Report

```
SQL> SELECT PRODUCT_NAME, PRICE, STOCK_QUANTITY,CASE WHEN STOCK_QUANTITY > 0 THEN 'AVAILABLE' ELSE 'OUT OF STOCK' EN
D AS AVAILABLILTY_STATUS FROM PRODUCT ORDER BY PRODUCT_NAME;

PRODUCT_NAME             PRICE STOCK_QUANTITY AVAILABLILTY
------------------- ---------- -------------- -----------
Bluetooth Speaker          1499             75 AVAILABLE
Building Blocks             999             35 AVAILABLE
Cooking oil                 699             70 AVAILABLE
Cotton Shirt                899             60 AVAILABLE
Data Structures             650             30 AVAILABLE
Denim Jeans                1599             45 AVAILABLE
Electric Kettle            1299             35 AVAILABLE
Face wash                   349             90 AVAILABLE
Office Chair               5499             15 AVAILABLE
Python programming          850             25 AVAILABLE
Running Shoes              2199             40 AVAILABLE
Smart phone               24999             50 AVAILABLE
Study Table                6999             10 AVAILABLE
Wireless Mouse              799            100 AVAILABLE
Yoga Mat                    599             80 AVAILABLE

15 rows selected.
```

Report 2: Customer Report

Total number of customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS FROM CUSTOMER;

TOTAL_CUSTOMERS
---------------
             15
```

New customer registrations

```
SQL> SELECT CUSTOMER_ID,CUSTOMER_NAME, EMAIL,REGISTRATION_DATE FROM CUSTOMER WHERE REGISTRATION_DATE >= DATE '2024-01-01' AND REGISTRATION_DATE < DATE '2026-01-01' ORDER BY REGISTRATION_DATE DESC;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                            REGISTRAT
------------------------------------------------------------------------------------------------ ---------
         15 Ishitha R
ishitha@gmail.com                                                                                10-SEP-25

         14 Gayathri R
gayu@gmail.com                                                                                   17-AUG-25

         13 Akash Kumar
akash@gmail.com                                                                                  16-JUL-25

         12 Suresh Babu
suresh@gmail.com                                                                                 25-JUN-25

         11 Ravi Chandran
ravi@gmail.com                                                                                   20-MAY-25

         10 Nithya S
nithya@gmail.com                                                                                 12-MAY-25

          9 Manoj R
manoj@gmail.com                                                                                  01-MAY-25

          8 Ajay Kumar
ajay@gmail.com                                                                                   10-APR-25

          7 Meeena K
meena@gmail.com                                                                                  02-APR-25

          6 Vijay Kumar
vijay@gmail.com                                                                                  08-MAR-25

          5 Anjali Devi
anjali@gmail.com                                                                                 01-MAR-25

          4 Karthik M
karthik@gmail.com                                                                                20-FEB-25

          3 Rahul Raj
rahul@gmail.com                                                                                  05-FEB-25

          2 Priya Sharma
priya@gmail.com                                                                                  15-JAN-25

          1 Arun Kumar
arun@gmail.com                                                                                   10-JAN-25

15 rows selected.
```

Report 3: Order Report

Total Orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS FROM ORDERS;

TOTAL_ORDERS
------------
          15
```

Completed Orders

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS FROM ORDERS WHERE ORDER_STATUS
 ='DELIVERED';

COMPLETED_ORDERS
----------------
               7
```

Pending Orders

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS FROM ORDERS WHERE ORDER_STATUS IN('PENDING','SHIPPING');

PENDING_ORDERS
--------------
             4
```

Cancelled Orders

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS FROM ORDERS WHERE ORDER_STATUS = 'CANCELLED';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

Highest-priced Products

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE FROM PRODUCT ORDER BY PRICE DESC;

PRODUCT PRODUCT_NAME              PRICE
------- -------------------- ----------
P001    Smart phone               24999
P014    Study Table                6999
P010    Office Chair               5499
P009    Running Shoes              2199
P005    Denim Jeans                1599
P003    Bluetooth Speaker          1499
P008    Electric Kettle            1299
P011    Building Blocks             999
P004    Cotton Shirt                899
P007    Python programming          850
P002    Wireless Mouse              799

PRODUCT PRODUCT_NAME              PRICE
------- -------------------- ----------
P012    Cooking oil                 699
P006    Data Structures             650
P015    Yoga Mat                    599
P013    Face wash                   349

15 rows selected.
```

Most Reviewed Products

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, COUNT(R.REVIEW
_ID) AS REVIEW_COUNT FROM PRODUCT P JOIN REVIEW R ON P.P
RODUCT_ID=R.PRODUCT_ID GROUP BY P.PRODUCT_ID,P.PRODUCT_N
AME ORDER BY REVIEW_COUNT DESC;

PRODUCT PRODUCT_NAME         REVIEW_COUNT
------- -------------------- ------------
P014    Study Table                     1
P003    Bluetooth Speaker               1
P012    Cooking oil                     1
P001    Smart phone                     1
P007    Python programming              1
P009    Running Shoes                   1
P015    Yoga Mat                        1
P013    Face wash                       1
P002    Wireless Mouse                  1
P004    Cotton Shirt                    1
P005    Denim Jeans                     1

PRODUCT PRODUCT_NAME         REVIEW_COUNT
------- -------------------- ------------
P006    Data Structures                 1
P008    Electric Kettle                 1
P010    Office Chair                    1

14 rows selected.
```

Available Products

```
SQL> SELECT PRODUCT_ID,PRODUCT_NAME, PRICE, STOCK_QUANTI
TY FROM PRODUCT WHERE STOCK_QUANTITY>0 ORDER BY STOCK_QU
ANTITY DESC;

PRODUCT PRODUCT_NAME              PRICE STOCK_QUANTITY
------- -------------------- ---------- --------------
P002    Wireless Mouse              799            100
P013    Face wash                   349             90
P015    Yoga Mat                    599             80
P003    Bluetooth Speaker          1499             75
P012    Cooking oil                 699             70
P004    Cotton Shirt                899             60
P001    Smart phone               24999             50
P005    Denim Jeans                1599             45
P009    Running Shoes              2199             40
P008    Electric Kettle            1299             35
P011    Building Blocks             999             35

PRODUCT PRODUCT_NAME              PRICE STOCK_QUANTITY
------- -------------------- ---------- --------------
P006    Data Structures             650             30
P007    Python programming          850             25
P010    Office Chair               5499             15
P014    Study Table                6999             10

15 rows selected.
```

Conclusion

In this practical, SQL queries were implemented on the existing E-Commerce Database to retrieve and analyze customer, product, order, and payment information. Basic SQL commands such as `SELECT`, `WHERE`, `ORDER BY`, and `DISTINCT` were used to display, filter, sort, and retrieve specific records.

Multiple filtering conditions and table joins were applied to obtain relevant information about products, customers, inventory, and orders. In addition, four business reports were generated to examine product availability, customer details, order status, and product performance.

Thus, this practical provided hands-on experience in using SQL queries to manage and retrieve meaningful information from an e-commerce database, supporting day-to-day operations and basic business analysis.